CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 16-2024: A 20-Year-Old Man with a Pustular Rash

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PRESENTATION OF CASE

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CME

Dr. Erica S. Shenoy: A 20-year-old man was evaluated in the emergency department of this hospital because of a diffuse pustular rash that had progressed over a period of several days.

Three days before the current evaluation, the patient noticed a pustular rash on the left forearm that stung while he was taking a shower. He was a student athlete and was evaluated by an athletic trainer, who referred him to the student health clinic. The next day, the rash had progressed to involve the right arm. He received prescriptions for oral trimethoprim–sulfamethoxazole and topical mupirocin.

The day before the current evaluation, the patient noticed an increased number of lesions, including new lesions on the chest, face, and ears. He consulted his primary care physician, who referred him to a dermatologist. The patient was unable to schedule a dermatology office visit, and the next day, he presented to the emergency department of this hospital.

In the emergency department, the patient reported that, despite topical and oral antimicrobial therapy, the rash had progressed and was very pruritic. He also stated that several days before the current evaluation, he had had close contact with another man with a similar rash. On a review of systems, he reported no fever, chills, night sweats, myalgia, cough, rhinorrhea, pharyngitis, nausea, vomiting, diarrhea, abdominal pain, or neurologic symptoms.

Medical history was notable for oral herpes simplex virus (HSV) infection, allergic rhinitis, patellar tendinosis, and tonsillectomy. Three years before the current evaluation, methicillin-sensitive *Staphylococcus aureus* infection involving the right arm had been treated with topical antibiotic agents. Current medications included oral trimethoprim–sulfamethoxazole and loratadine and inhaled albuterol as needed. He had no known drug allergies. He had received two vaccinations against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) but had not received the vaccine against mpox because he had no risk factors for infection. There was no family history of infectious, dermatologic, rheumatologic, or oncologic conditions. The patient was a college student and lived in a dormitory on campus; he was on the college's wrestling team. Previously, he had lived in a town in coastal Massachusetts but reported no other travel. He had a pet dog, but there were no other animal exposures. He was sexually active with women. He had a history of vaping but did not drink alcohol, smoke tobacco, or use illicit substances.

On examination, the temporal temperature was 36.2°C, the heart rate 88 beats per minute, the blood pressure 129/74 mm Hg, the respiratory rate 18 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. He was muscular; the weight was 77 kg. He had dozens of grouped, raised vesiculopustular lesions with erythematous bases at various stages of development; some of the lesions were more pustular, some were umbilicated, and others had a scab or eschar. Some areas of confluence were present. Lesions were found on the antecubital fossae, cheeks, and torso; similar lesions were scattered on the back, left axilla, lower right abdomen, calves, occiput, and posterior to the ears (Fig. 1). There was erythema in intertriginous areas. The remainder of the examination was normal.

The blood levels of electrolytes, glucose, albumin, globulin, alanine aminotransferase, aspartate aminotransferase, and bilirubin were normal, as were the results of kidney-function tests; other laboratory test results are shown in Table 1. Tests for SARS-CoV-2 and human immunodeficiency virus (HIV) types 1 and 2 were negative.

Diagnostic tests were performed, and management decisions were made.

Table 1. Laboratory Data.

Variable	Reference Range, Adults*	On Initial Evaluation, This Hospital
Hemoglobin (g/dl)	13.5–17.5	15.3
Hematocrit (%)	41.0–53.0	46.8
White-cell count (per μl)	4500–11,000	4650
Differential count		
Neutrophils (per μl)	1800–8100	2390
Lymphocytes (per μl)	1200–5200	1150
Monocytes (per μl)	200–1400	740
Eosinophils (per μl)	0–1000	210
Immature granulocytes (%)	0–0.9	1.5, including metamyelocytes, myelocytes, and promyelocytes
Platelet count (per μl)	150,000–400,000	198,000

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

Figure 1. Clinical Photographs.
Skin lesions are present on the right side of the face (Panel A), on the right side of the neck and behind the right ear (Panel B), on the left antecubital fossa and axilla (Panel C), and on the right antecubital fossa (Panel D).

DIFFERENTIAL DIAGNOSIS

Dr. Demetre C. Daskalakis: This patient is a 20-year-old male athlete who presents with a vesiculopustular rash, the clinical feature around which I will build my differential diagnosis. Although it is important to consider both infectious and noninfectious causes of this rash, infection seems to be the most likely cause, given the patient's history and examination findings.

NONINFECTIOUS CAUSES OF VESICULOPUSTULAR RASH

I will start with the less likely noninfectious causes of this patient's rash. Autoimmune diseases such as sarcoid, various perforating dermatologic disorders, and porphyria should be considered, although there are no clues in the patient's presentation that would support these diagnoses. Certain types of cancer, particularly leukemia cutis or other hematologic cancers, could also involve a vesiculopustular rash, but this patient appears to be otherwise well. Drug reactions should also be considered, given the use of a sulfa-containing antimicrobial agent, as should erythema multiforme. Other possible noninfectious causes include neutrophilic dermatoses (e.g., Sweet's syndrome), vesicular pityriasis rosea, and eczema vaccinatum.

INFECTIOUS CAUSES OF VESICULOPUSTULAR RASH

The patient's vesiculopustular rash is most likely caused by an infectious process. Infection with viruses, bacteria, and fungi, as well as parasitic infestations, should be considered. Possible viral causes include infection with herpesviruses (e.g., HSV type 1 [HSV-1] and type 2 [HSV-2] and varicella–zoster virus [VZV]) and poxviruses (e.g., molluscum contagiosum virus and mpox virus [MPXV]). Other viruses that are less likely to be the cause of this patient's rash include parvovirus, enterovirus, measles virus, and HIV. Possible causes involving bacterial infections include community-acquired *S. aureus*–associated folliculitis and septicemia, pseudomonas folliculitis, rickettsia infection (especially rickettsialpox), anthrax, nontuberculous mycobacterial infection, syphilis, and scrub typhus. Fungal infections — including cryptococcosis, histoplasmosis, and talaromycosis — could cause a rash similar to that in this patient. Scabies infestation with a nodular presentation should also be considered.

The patient's presentation is extremely mild with no prodromal or concurrent systemic symptoms, characteristics that make many of these diagnoses unlikely. He participates in wrestling, a sport that involves persistent skin-to-skin contact; he reported having close contact with another person with a similar rash, which increases the likelihood of an infection associated with such contact. The rash progressed in a centripetal distribution (from the arms to the core) rather than a centrifugal distribution (from the core outward), which further narrows the differential diagnosis, including making measles less likely than other causes. The description of multiple lesions at different stages of evolution — some more pustular, some umbilicated, and some with a scab or eschar — further helps to refine the differential diagnosis. In addition, the absence of lymphadenopathy is a clinically significant finding that also helps to narrow the list of possible causes.

The absence of an exposure history makes infection with nontuberculous mycobacteria, rickettsia, endemic fungi, HIV, measles virus, and *Treponema pallidum* (which causes syphilis) unlikely. Eczema vaccinatum seems improbable, given that it is unlikely that the patient had a recent smallpox vaccination. The mild clinical presentation and course make infection with more pathogenic organisms such as *Bacillus anthracis* (which causes anthrax), certain fungi, or certain bacteria unlikely. The fact that the patient is immunocompetent makes infections such as cryptococcosis, talaromycosis, and histoplasmosis, as well as nodular scabies, unlikely. *S. aureus*–associated and pseudomonas folliculitis, erythema multiforme, and drug-induced rash are also unlikely diagnoses, given the morphologic features and pattern of the observed rash. Although staphylococcal infections can occur during close skin-to-skin contact in the community, the appearance of this patient's skin lesions is not suggestive of this type of infection despite his history of such infection.

The presence of promyelocytes on the automated differential count is a concern and should be investigated with a follow-up manual differential count; however, this finding is probably not indicative of a hematologic cancer, given the absence of other clinical findings. HSV-1 infection manifesting as herpes gladiatorum or eczema herpeticum, primary VZV infection or reactivation of VZV, and poxvirus infections such as mpox or molluscum contagiosum remain on the list of possible diagnoses in this patient.

HSV Infection

Outbreaks of HSV-1 infection that are caused by close physical contact in the context of wrestling, also known as herpes gladiatorum, have been reported and are not infrequently observed.[1] This diagnosis is suggested by the presence of vesicular lesions in wrestlers in body areas that make contact with their opponents. The distribution of the lesions is classic and correlates with some of the common skin exposures expected in wrestling maneuvers while the wrestler is wearing a wrestling singlet. In this patient, the rash started on the arms and is present almost entirely on skin surfaces that are exposed during wrestling. The patient's report of close physical contact with someone with a rash similar to his rash further supports this diagnosis. The finding that the rash is in multiple stages of development is also consistent with infection with other herpesviruses, including VZV in particular. Eczema herpeticum — disseminated HSV-1 infection in a patient with underlying atopic dermatitis — is another possible cause of this patient's rash on the basis of the presence of lesions on extensor surfaces and the patient's history of allergic rhinitis. However, the absence of a known underlying skin disease, along with the described distribution of the rash, makes this diagnosis unlikely. Although the patient's history of oral HSV infection may be protective against these HSV-1–associated manifestations, previous reported outbreaks of HSV-1 infection among wrestlers indicate that this protection is, at best, incomplete.[2]

VZV Infection or Reactivation

VZV infection or reactivation should remain on the list of possible causes of the vesiculopustular rash in this patient because of its morphologic features, with lesions described to be in crops at various stages of evolution. The lack of a clear history of chickenpox or vaccination in the information we are given may also increase the possibility of primary VZV infection. Features that make primary infection less likely than other causes include the absence of systemic symptoms, the limited dissemination of the rash (which is located primarily in areas of exposed skin), the absence of prodromal symptoms, and the centripetal (rather than centrifugal) pattern of spread. Shingles, or VZV reactivation, should also be considered because of the morphologic characteristics of the rash, but the fact that the rash was not in a dermatomal distribution makes this diagnosis less likely than others. Disseminated reactivation of VZV is also less likely, given the absence of underlying immunocompromise.

Poxvirus Infection

The other infectious pathogens that remain high on my differential diagnosis are poxviruses, specifically MPXV.[3] This virus has been known to cause clinically significant human disease in Africa for decades, with exportations occasionally leading to limited human-to-human spread outside of areas in which mpox is endemic; a large global outbreak began in 2022. This outbreak was epidemiologically associated with sexual contact, and the vast majority of reported cases were in men who have sex with men.[4,5] Although intimate contact was often reported in association with this outbreak, MPXV can be transmitted through any skin-to-skin contact and is less commonly transmitted through contact with fomites such as linens. For this reason, mpox should be considered in this patient even though he does not report sexual contact with other men and does not have any of the mucosal manifestations that have frequently been reported in the current global outbreak.

The patient's social history — although not typical of that reported in the current outbreak — describes close physical contact that may support the acquisition of MPXV infection. The vesiculopustular morphologic features of this rash (which include umbilication), along with the recent global epidemiologic characteristics of this outbreak, support the possibility of mpox. However, mpox lesions tend to be observed in the same — rather than different — stages of evolution, which makes the presence of MPXV less likely than that of a herpesvirus in this patient. Classically, mpox is preceded by a febrile prodrome, which this patient did not report, but such a prodrome was not commonly described in the current global outbreak. The absence of lymphadenitis associated with the rash also makes mpox unlikely. Molluscum contagiosum should also be considered because of the morphologic features of the rash, the mechanism of transmission, and the absence of lymphadenopathy.[6] However, the observation of lesions at multiple stages of development, along with the large number of lesions in this immunocompetent host, make this diagnosis less likely than others.

Given the constellation of clinical findings, the morphologic characteristics and progression of the rash, and the exposures described by this patient, I suspect that herpes gladiatorum caused by HSV-1 infection is the most likely diagnosis. To establish the diagnosis, I would obtain a swab of one of the lesions for HSV-1 nucleic acid testing.

CLINICAL IMPRESSION

Dr. Howard M. Heller: The evening that this patient presented to the emergency department, I was on call for our biothreats response team — the team that responds to communicable highly contagious diseases such as Ebola virus disease and mpox in the context of the initial outbreak. Our hospital system had implemented a real-time electronic decision-support tool to help clinicians determine whether a patient warrants evaluation for mpox, and this patient met those criteria.[7]

When I learned that this patient was a college wrestler with vesicular lesions, my first thought was that he had herpes gladiatorum. However, when I saw the rash, I became more concerned because it was far more extensive than I had ever seen in a patient with herpes gladiatorum. This patient presented near the height of the mpox outbreak, and we were seeing cases of mpox in persons who did not have identifiable risk factors,[8] but this patient had several teammates with rashes similar to his. Considering the potential infection-control ramifications, I was concerned about mpox.

I spoke with the patient's coach, who informed me that several teammates had recently had rashes that had been treated by the team doctor with trimethoprim–sulfamethoxazole and that the rashes had resolved. I also learned that the evening that the patient presented was the opening weekend of wrestling season, and his team was in another state for a match. Until that point, he had been practicing with only his sparring partner.

The decision to evaluate for mpox could potentially preclude other testing because of the restrictions for specimen processing, including the processing of bacterial cultures and samples for HSV nucleic acid amplification testing. I contacted the Massachusetts Department of Public Health (MDPH) and the medical director of the special pathogens program at this hospital to expedite testing for mpox and HSV infection.

CLINICAL DIAGNOSIS

Herpes simplex virus infection resulting in herpes gladiatorum.

DR. DEMETRE C. DASKALAKIS'S DIAGNOSIS

Herpes simplex virus infection resulting in herpes gladiatorum.

INFECTION-CONTROL RESPONSE

Dr. Shenoy: I was the infectious disease consultant who saw the patient the morning after the current evaluation and also addressed the infection-control aspects of the case. Although several possibilities were included on the differential diagnosis, the infection-control considerations were dominated by the possibility of mpox.

Infection-control management of suspected or confirmed mpox involves standard precautions, with additional measures including the placement of a patient in a single-patient room with a dedicated bathroom or commode and the use of specific personal protective equipment by health care personnel. Such equipment includes a gown, gloves, eye protection, and an N95 respirator or higher-level protection. Special air handling (i.e., the use of an airborne-infection isolation room) for patients with suspected or confirmed mpox is not necessary; however, procedures such as intubation that are likely to generate aerosols should preferentially be performed in an airborne-infection isolation room.

One of the key aspects of infection prevention and control is early identification of relevant presenting signs, symptoms, and epidemiologic findings in a patient, followed by appropriate isolation and communication with relevant infection-control and public health authorities — the identify–isolate–inform approach that has been applied to several emerging infectious diseases.[9]

In this case, the clinical team caring for the patient used a clinical decision support system that was developed at this hospital and embedded within the electronic health records to guide clinicians through the evaluation of patients presenting with signs and symptoms of mpox and relevant exposure. This support system allowed for the automatic initiation of appropriate isolation of a patient, guided diagnostic evaluation, supported empirical treatment (on the basis of patient eligibility), and connected to infection-control personnel within the hospital. The clinical decision support system was deployed at the start of the mpox outbreak[10]; a recent analysis of 668 uses of the system documented a positive predictive value of 35% and a negative predictive value of 99%.[7] Given that mpox was on the differential diagnosis, along with the possibility of other cases involving teammate contacts, we were in close communication with colleagues at the MDPH. While the diagnostic results were pending, the patient received a prescription for tecovirimat as part of an expanded-access protocol for the empirical treatment of possible mpox, as well as a prescription for valacyclovir for the empirical treatment of herpesvirus infection.

PUBLIC HEALTH ASSESSMENT AND DIAGNOSTIC TESTING

Dr. Katherine Hsu: The MDPH was contacted and became immediately involved. Nucleic acid testing for nonvariola orthopoxviruses (including MPXV) was coordinated through the state public health laboratory of the MDPH, which offered the fastest turnaround time at that point in the mpox epidemic. Given the appearance of this patient's lesions, it was also important to evaluate for alternative diagnoses. A specimen was obtained from a lesion, and with our input, the hospital team arranged for HSV-1, HSV-2, and VZV polymerase-chain-reaction testing to be performed at the microbiology laboratory at Boston Medical Center, which had available weekend testing.

In parallel, epidemiologists at the MDPH began an investigation to identify any symptoms of illness or rash in the patient's recent contacts. This process involved interstate notification and coordination, since the patient's teammates were away at a wrestling meet in a neighboring state. When the patient was discharged, he was provided with instructions to remain in isolation at home until his test results became available.

By the next day, the epidemiologists had confirmed that the patient's teammates who had been traveling had no symptoms of mpox. The patient's tests were found to be negative for nonvariola orthopoxvirus, HSV-2, and VZV but positive for HSV-1. These results confirmed the diagnosis of HSV-1 infection and obviated the need to issue guidance regarding mpox-specific quarantine. Treatment with tecovirimat was discontinued, and valacyclovir therapy was continued.

DISCUSSION OF DIAGNOSIS AND MANAGEMENT

Dr. Heller: An early description of an HSV outbreak among wrestlers was reported in 1964, and the term herpes gladiatorum was coined.[11] Since then, such infections have been well described among wrestlers, as well as among participants in other sports such as rugby (herpes rugbiorum). A notable outbreak occurred in a high school wrestling camp in Minnesota in 1989 in which 60 campers were affected.[1]

Transmission is thought to occur either by direct (skin-to-skin) contact or through secretions on the wrestling mat with which the athletes have contact. Close contact may be prolonged in wrestling because of the nature of the sport. Autoinoculation can occur. Spread of lesions, in multiple stages of development, can also occur. Lesions tend to appear primarily on the face, the sides of the neck, and the forearms. Guidelines recommend treatment with valacyclovir twice daily for 10 to 14 days for primary infection and twice daily for 5 to 7 days for recurrent infection.[12]

Several strategies have been suggested for the prevention of HSV-1 transmission. Education for athletes, trainers, and coaches is essential for recognition of the signs and symptoms, understanding the mode of transmission, and implementation of preventative measures. One such measure involves cleaning mats with a virucidal agent. In a clinical trial that was conducted at a wrestling camp for teenagers, those who were HSV-1 seropositive were treated with valacyclovir prophylactically and those who were not seropositive did not use any prophylactic antiviral medication. The use of valacyclovir resulted in an 87% lower incidence of herpes gladiatorum than no use of antiviral medication during the 28-day observation period.[13]

Guidelines specific to both high school and college wrestlers regarding the duration of antiviral therapy and the crusting of lesions before resumption of participation in wrestling are available. For prophylaxis, a 1-g daily dose of valacyclovir is recommended if the most recent infection occurred less than 2 years earlier, and a 500-mg daily dose of valacyclovir is recommended if the most recent infection was more than 2 years earlier.

To our knowledge, no cases of mpox resulting from contact sports have been reported, but the potential for outbreaks exists. Guidelines for isolation and daily assessment of exposed athletes that are based on the standard mpox guidelines have been proposed.[14] Education for athletes, trainers, coaches, massage therapists, and others who have close contact with athletes should include mpox in addition to their standard training that includes *S. aureus* infection and other communicable diseases. Sports medicine staff should have plans in place for evaluating suspected cases.

One week after the patient's initial presentation to this hospital, he was seen at his college health center; all the existing lesions had scabbed, and no new lesions had developed. It was recommended that he continue to take valacyclovir prophylactically for the duration of the wrestling season. Prophylactic treatment with valacyclovir had been recommended and offered to the wrestling team at the start of the season, although not all the athletes, including this patient, had taken it.

FINAL DIAGNOSIS

Herpes simplex virus infection resulting in herpes gladiatorum.

The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the Centers for Disease Control and Prevention.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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